

- * Compact central regions: emits copious amounts of non-stellar light
 - Powered by accretion onto a SMBH

AGN: Active galactic Nuclei:

- Star light: UV: $3000 \text{ \AA}$
100 μ far IR
- Others have strong emission at
 (i) gamma ray: $\lambda_g \sim 10^{-11} \text{ m}$, $E \sim 10^5 \text{ eV}$
 (ii) Radio: $\lambda_r \sim 10 \text{ MHz} - 100 \text{ GHz}$ $E \sim 10^{-7} \text{ eV}$

* Activity: 1% of galaxies in the local universe
 ◦ "low-level" activity" $\sim 50\%$.

- Magnetic fields: Reconnection & Parker / Dynamo theory
 → Shocks in ISM

Seyfert Gal:

- first AGN (lowest L) $L \sim 10^{35} - 10^{38} \text{ W}$
- Bright nuclei of nearby spirals
- 1%.
- Type I: narrow & broad emission lines
 Type II: narrow lines
 Type I: bluer

QSO

Quasars (Most luminous) $10^{39} \sim 10^{42} \text{ W}$

- taking optical spectra of extragalactic radio sources
- Radio quiet
- Point-like: blue luminaries

Exceeds L_{gal} by $L_Q \sim 3(L_{gal})$

$> 10^6$ known

$z = 7.64$ (furthest)

X-rays: "corona" of accretion disk

Corona $\sim 10^4 K$

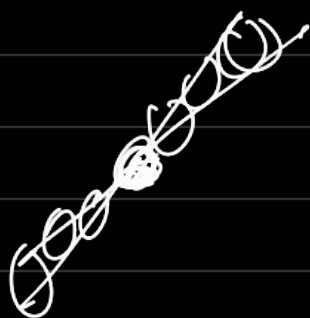
Radio Galaxies:

• Ellipticals

• HERG: High excitation radio gal:

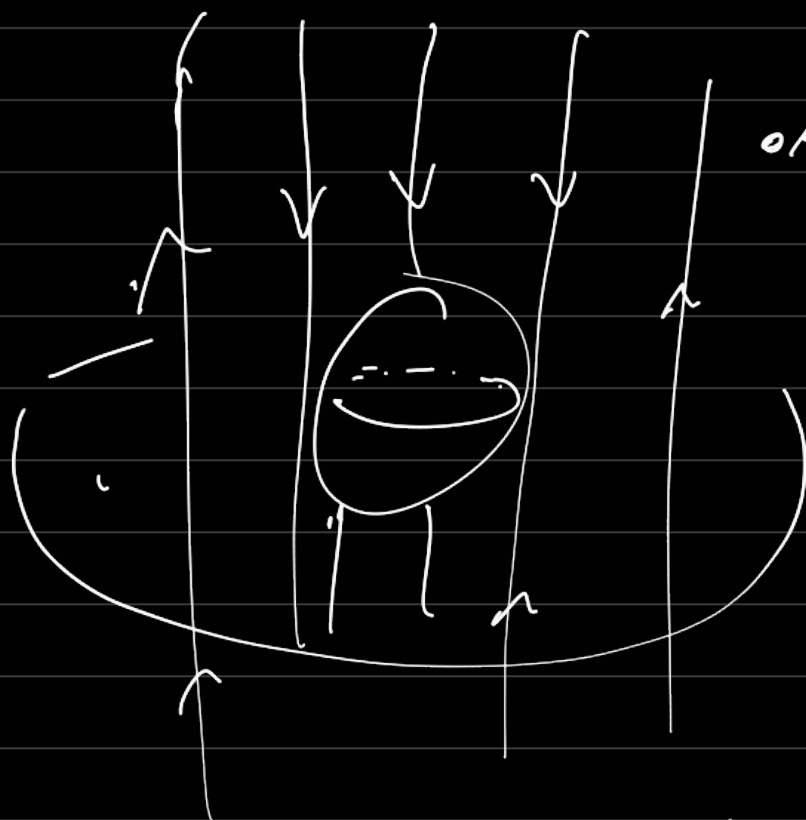
Type II

LERG: weaker optical emission lines

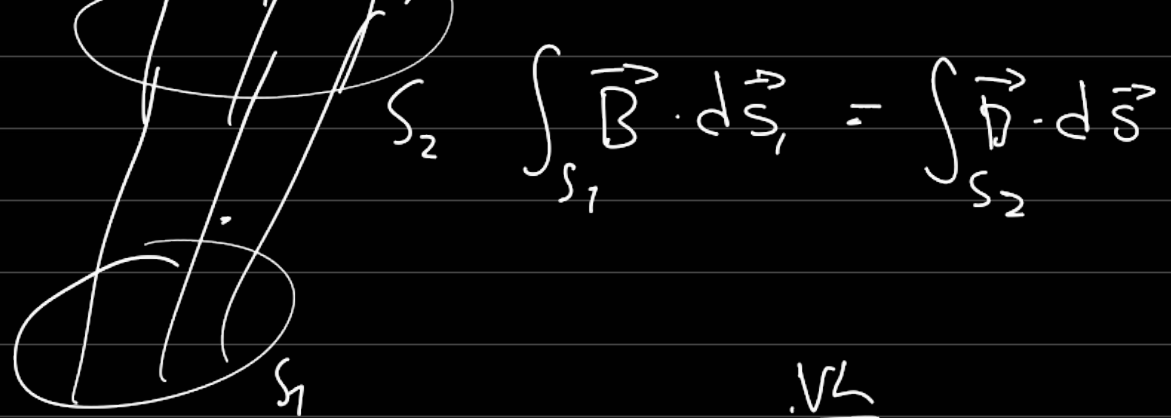


MHD

• Alfvén waves

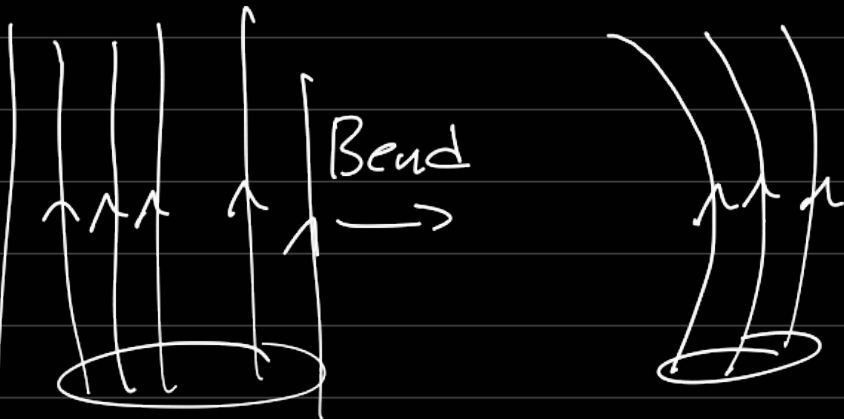


Alfvén flux freezing: $\frac{\partial \vec{B}}{\partial t} \approx \nabla \times (\vec{v} \times \vec{B})$



* Magnetic Reynolds #: $R_m \sim \frac{VL}{\eta}$

Alfvén's theorem of flux freezing: ↗



→ Tells us: in astrophysical systems ($L \gg 1$) we can almost regard plasma as a "material" which we can bend, twist or distort

* Lab: $\frac{\partial \vec{B}}{\partial t} \sim \eta \nabla^2 \vec{B}$

* In Astro:

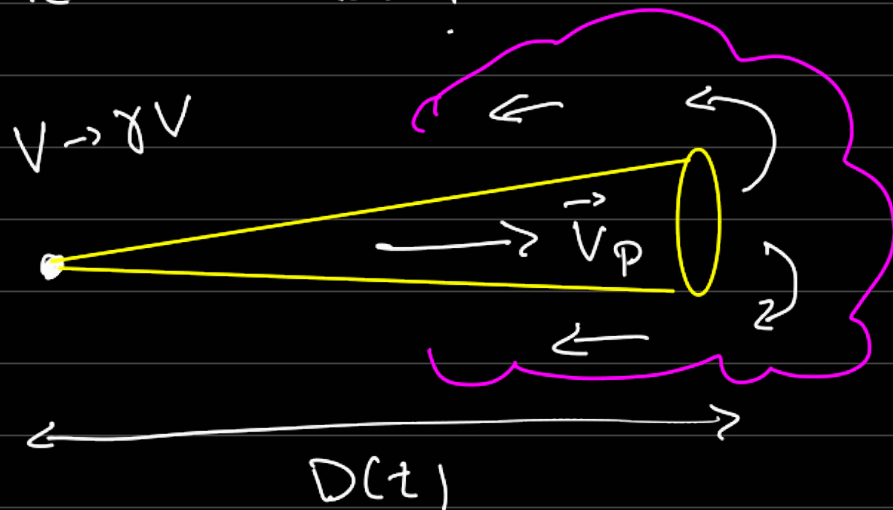
(i) initial conf. & info of plasma flow
 → Allows us to guess mag field conf.
 By flux freezing

NLR: (Narrow - line - regions)

- Receive radiation only from gas blobs much further away from central region (more slowly)

BLR: Broad $\rightarrow$ closer

FR II

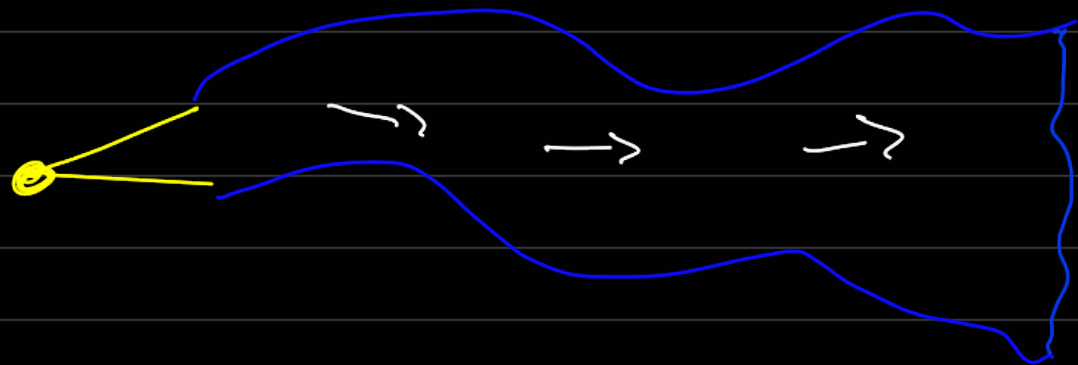


[Jet
Lobe

(Double radio gal.)

- Constant opening angle $\theta \leq \theta$
- Propagates into external medium
- Length of jet $D(t)$ does not grow as fast as $\vec{V}_p$
i.e. $|D'(t)| < |\vec{V}_p|$
- Plasma will "hit" external medium cause shock and spread out
- Shock compresses rel. particles $\rightarrow$ strong emission source.

FR I



Jet

Tail

- Jet rel. collimated, soon/quicker destabilizes

- Long Diffuse tails
 - Much more common
 - Much more complex
- Subsonic tails
 - Sensitive to local IGM
 - Brighter near center in jet via syn.

- Pair production:
 - Inverse Compton

AGN's: $F_{\nu} \propto \nu^{-\alpha}$

- More prominent the further we look out
 - $\sim z=2$ already yields ≥ 100 more than in our own world

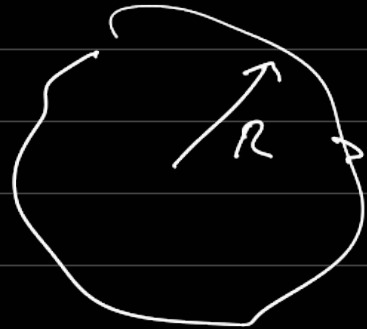
BH accretion:

- Crossing time Δt

$$R \leq c \Delta t$$

$$R \leq 0.1 \text{ pc} : \text{observations radio-opt.}$$

$$R \leq 10^5 \text{ pc} : \text{Y-ray}$$



R : comparable to the Schwarzschild Radius

$$R_{\text{Sch}} = \frac{2GM_{\text{BH}}}{c^2}$$

$$K = \frac{1}{2} mc^2 = \frac{GM_{\text{BH}}}{R}$$

- $K_{sc} = c^2$
- No other system can have the en. density
→ only BH.

Quasar: $L = 10^{40}$ W

- Nuclear burning in stars:

$$E = \epsilon mc^2 \quad (\epsilon \approx 0.7\% = 0.007)$$

- Via Nuclear fusion: $250 M_{\odot}/\text{yr}$
- Not the correct mechanism:

$$\Delta U = U_i - U_f = 0 - \left(-\frac{GMm}{R_{\text{sch}}} \right) = \frac{GMm}{R_{\text{sch}}}$$



- Suppose all ΔU can be radiated

$$E = \epsilon mc^2 = \frac{GMm}{R_{\text{sch}}} \quad \left| \quad \epsilon = \frac{GM}{R c^2} \right.$$

$$\epsilon_{\text{max}} = \frac{GM_{\text{B.H.}}}{R_{\text{sch}} c^2} = \frac{c^2 GM_{\text{BH}}}{2GM_{\text{B.H.}} c^2} = \frac{1}{2}$$

$$\epsilon_{\text{max}} = \frac{1}{2}$$

- Not all material falls in: Most collects in orbits

Mechanism: Falling particles:

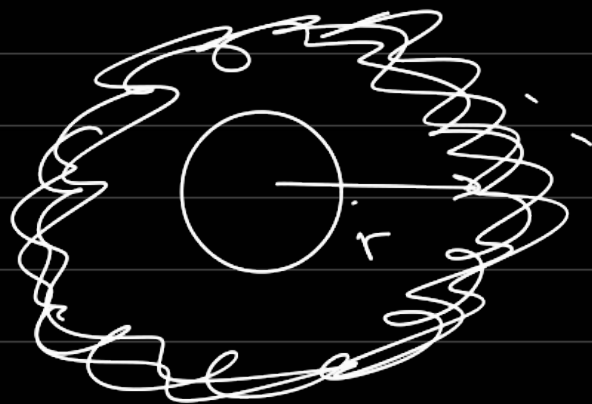
$$(\text{grav. Energy}) \longrightarrow (\text{heat}) \oplus (\text{Radiation})$$

- Main component of luminosity

* Aside: Penrose - Process

- Extracts Energy from B.H.

- Energy available from the Accretion disk



- Mass m in orbit (Assume Kepler)

$$* U = -\frac{GMm}{r} \quad | \quad K = \frac{GMm}{2r}$$

$$\bullet \frac{GMm}{r^2} = \frac{mV^2}{r} \Rightarrow V^2 = \frac{GM}{r}$$

$$K = \frac{1}{2} m V^2 = \frac{1}{2} m \left(\frac{GM}{r} \right) = \frac{GMm}{2r}$$



$$\Delta U = \Delta r \frac{dU}{dr} = \Delta r F = \Delta r \frac{GMm}{r^2}$$

(Potential energy lost)

$$|\Delta K| = \Delta r \left| \frac{dK}{dr} \right| = \Delta r \frac{GMm}{2r^2} = \frac{1}{2} \Delta U$$

- 50% of released grav. potential goes to kinetic

50% goes into heating

=> Matter from the accreting disk

$$\epsilon_{\max} = \frac{1}{4}$$

Since $E = \frac{1}{2} \Delta U = \epsilon m c^2$

$$\epsilon = \frac{\frac{1}{2} G M}{r c^2} \quad r = R_{\text{sch}}$$

$$\dots (\epsilon_{\max})|_{R \rightarrow \infty} = \frac{1}{4}$$

* Schwarz: last stable orbit: $3 R_{\text{sch}}$

$$\epsilon_{\max} \approx \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12} \approx 8\%$$

* Rotating B.H.: $\epsilon \approx 42\%$

→ standard $\epsilon = .1$ (10%)

$$L = \epsilon \frac{dE}{dt} = \epsilon \dot{M} c^2$$

$$E = m c^2$$

$$L = \epsilon \dot{M} c^2$$

Maximum Accretion Luminosity: Eddington Limit

- Radiation from Nucleus: exerts outward force
- Counter balanced by inward grav.
- When balanced: No net force and net influx stops

$$P = E/c$$

$$F = \frac{dP}{dt} = \frac{1}{c} \frac{dE}{dt} = \frac{1}{c} L_r = \frac{1}{c} \frac{\sigma_T}{4\pi r^2} L$$

$$\int \Delta F_{\text{rad}} \approx \frac{1}{c} \frac{\Delta E}{\Delta t}$$

$$v = \frac{1}{t} \Rightarrow \Delta N = - \frac{\Delta t}{t^2} = - \frac{\Delta t}{v^2}$$

$$|\Delta t| = v^2 \Delta N$$

$$\frac{\Delta E}{\Delta t} = L$$

$$\text{but } L \Delta t = L v \Delta N$$

$$L v \Delta N = L v \Delta N$$

$$L v = L v^2$$

$$\frac{1}{c} L v \frac{\Delta N}{\Delta t}$$

$$F_{\text{rad}} = \int_0^\infty \frac{L v}{h v} \frac{h v}{c} dN$$

$$F_{\text{rad}} = \frac{\sigma_T}{4\pi R^2} \int_0^\infty \frac{L v}{h v} \frac{h v}{c} dN$$

$$F_{\text{rad}} = \frac{\sigma_T L}{4\pi c R^2} \quad (\text{force on an electron at } R)$$

$$\bullet \text{ Grav: } F_{\text{grav}} = \frac{G M (m_e + m_p)}{R^2} \approx \frac{G M m_p}{R^2}$$

$$F_{\text{grav}} = F_{\text{rad}} \Rightarrow L_{\text{max}} = L_{\text{Edd}} = \frac{4\pi G M_{\text{BH}} m_p c}{\sigma_T}$$

$$L_{\text{Edd}} \approx 1.26 \times 10^{31} \left(\frac{M_{\text{BH}}}{M_\odot} \right) \text{ W}$$

* Maximum Luminosity of a source of mass M which is powered by spherical Accretion
(Eddington - Limit)

$$L = \dot{E} c^2 = \frac{4\pi G M_{\text{BH}} m_p c}{\sigma_T}$$

• $\dot{M}$:

$(1-\epsilon)\dot{M}$: radiation

$\epsilon\dot{M}$: gravitation ('physical' mass)

$$\frac{dM_B}{dt} = (1-\epsilon) \frac{dM}{dt}$$

$$\frac{\epsilon}{(1-\epsilon)} \dot{M}_{BH} c^2 = \frac{4\pi G M_{BH} \rho c}{\sigma_T}$$

$$\int \frac{dM_{BH}}{M_{BH}} = \int \frac{(1-\epsilon) 4\pi G M_{BH} \rho c}{\epsilon \sigma_T} dt$$

$$\ln(M_{BH}) = kt + C$$

$$M_{BH} = M_0 e^{kt}$$

• Doubles mass in $\approx 3 \times 10^7 \text{ yr}$
